

Supplementary Information

Auditable multimodal integration of metagenomic sequencing and clinical evidence
identifies causative pathogens in lower respiratory tract infection

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Supplementary Figures

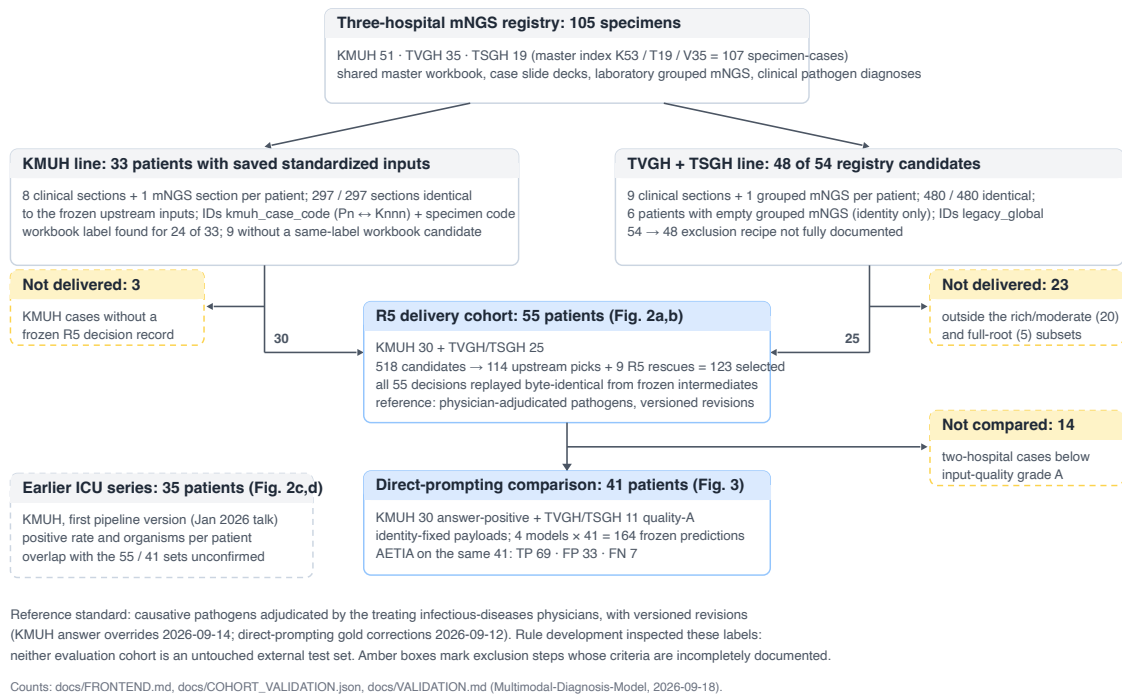


Figure S1: **Cohort flow.** From the three-hospital mNGS registry (105 specimens; Table 1) through the two hospital lines to the R5 delivery cohort ($n = [55]$) and the direct-prompting comparison cohort ($n = [41]$). Amber boxes, exclusion steps whose criteria are incompletely documented in the source materials; dashed grey box, the earlier 35-patient intensive-care series (main Fig. 2c,d), whose overlap with the evaluation cohorts is [unconfirmed]. Counts from the repository's cohort validation of 2026-09-18.

Supplementary Tables

Table S1: **mNGS organism features computed before scoring.** Source: `upstream/tools/rank_mngs_candidate_microbes.py`, `upstream/tools/mngs_common.py`, `deterministic_mngs_max_scorer.py` (`dominance_by_record`).

Feature	Values	Definition
Control-subtracted priority	1–5	1: NTC=00 and RK00; 2: RK8 and NTC=00; 3: RK00; 4: RK8; 5: NTC=00 only. Code score 1.0, 0.85, 0.65, 0.45, 0.30.
Read percentile	0–1	Percentile of $\log_{10}(\text{reads} + 1)$ among all organisms in the run.
Final score / label	high, medium, low	$0.6 \times \text{code score} + 0.4 \times \text{percentile}$; high ≥ 0.70 , medium ≥ 0.45 .
Read tier	R0–R4	R0: 1–9 reads; R1: ≥ 10 ; R2: ≥ 100 ; R3: $\geq 1,000$; R4: $\geq 10,000$.
Dominance tier	D0–D3	Per specimen record; top organism: D3 if reads $\geq 10 \times$ runner-up, D2 if $\geq 3 \times$, else D1; every other organism D0.
Specimen align-ment	aligned, sterile/systemic, non-aligned	From the mNGS-to-specimen assignment and the target site (lower respiratory tract).
Hospital support	per module	Same-organism support from culture, FilmArray/galactomannan, molecular microbiology; “Support”, “Not support” or “Unknown”.
Host context	V0–V3, O0–O2	Host-vulnerability tier and opportunistic-coverage level; host expansion when V2–V3 or O1–O2.

Table S2: **Signal tiers and causative levels of the deterministic scorer (v20).** Source: `deterministic_mngs_max_scorer.py`, functions `m1_strong`, `m2_moderate`, `m3_protected`, `m4_weak`, `set_candidate_level_from_tier`, `level3_pickable`, `pickable`. Conditions are evaluated in order; the first satisfied tier applies.

Tier / level	Condition
M1 strong	priority 1–2 and R3–R4; or priority 1–2 and percentile ≥ 0.85 ; or D3 and $\geq R2$ and priority ≤ 3 .
M2 moderate	priority 1–2 and R2 and percentile ≥ 0.75 ; or D2–D3 and $\geq R2$ and priority ≤ 3 ; or same-organism auxiliary support and $\geq R1$ and priority ≤ 3 ; or protected pathogen (not reactivation virus or mould), $\geq R1$, priority ≤ 3 , label not low; or R3–R4 and priority ≤ 4 and label not low; or <i>P. jirovecii</i> with R1–R2 and priority ≤ 3 ; or Mucorales at priority 1 with percentile ≥ 0.60 . Reactivation viruses and <i>Aspergillus</i> use dedicated clauses.
M3 protected	retention condition of a protected pathogen; or priority 1–2 with R0–R2; or priority ≤ 3 with percentile ≥ 0.60 above R0; or priority ≤ 3 with high/medium label and R1–R2; or priority 3 with R1–R2 and non-host auxiliary support; or priority 4 with R4.
M4 weak	priority 4–5; or label low; or R0–R1; or non-aligned specimen.
M5 background	organism on the background list.
Level 1	M1, aligned or sterile specimen, and (D3 or R3–R4).
Level 2	other M1; M2 with aligned or sterile specimen.
Level 3	M2 non-aligned; M3; M4 under host expansion when plausible (not reactivation virus, colonizer or non-aligned; with non-host support, protected status, or priority ≤ 3 with R1 and high/medium label).
Level 4	other M4; any organism demoted by a guardrail.
Level 5	M5.
Picked	level 1–2; level 3 with protected status, an oral-flora/respiratory-commensal representative rule, related typical-pneumonia context, direct support from culture, FilmArray/galactomannan or molecular testing, or (host expansion) priority ≤ 2 with $\geq R2$ or D2–D3; <i>Candida</i> /generic yeast never at level 3 without invasive support. Fallback: level-4 “best available” when nothing is pickable.

Table S3: **Guardrail rule identifiers of the deterministic scorer.** All 33 R-S4-* identifiers found in `deterministic_mngs_max_scorer.py`. Descriptions paraphrase the identifier and the rule body; [\[verify each against the code before submission\]](#).

Identifier (R-S4-...)	Effect
BACKGROUND	background organism, level 5
CANDIDA-BLOOD-STERILE-CONTEXT	<i>Candida</i> retained on blood/sterile-site evidence
CANDIDA-HOSPITAL-NONINVASIVE-NOT-PICKED	<i>Candida</i> with only non-invasive hospital support not picked
CANDIDA-RESP-CONTEXT-NOT-PICKED	respiratory <i>Candida</i> context not picked
CANDIDA-RESP-L3	respiratory <i>Candida</i> held at level 3; picked only with invasive support
COMMON-HOSPITAL-R0R1-NONDOMINANT-PICKED-OMIT	common hospital organism with R0–R1, non-dominant: omitted from picks
GENERIC-UNIDENTIFIED-HOSPITAL-LABEL	generic labels (e.g. “Gram-negative bacilli”) never picked
HERPES-HOST-R2R3-NONDOMINANT-REVIEW	herpesvirus with host-only support, R2–R3, non-dominant: review only
HERPES-R4-NONDOMINANT-HOST-ONLY-REVIEW	herpesvirus with R4 but non-dominant and host-only: review only
IMPOSSIBLE-INFECTION-SOURCE	organism cannot be a pulmonary infection source
L3-MNGS-ONLY-HOST-LOW-NONDOMINANT	level-3 mNGS-only organism with low, non-dominant reads not picked
LOW-ACTIONABILITY-VIRUS-PICKED-OMIT	EBV, HHV-6/7 and similar omitted from picks
LOW-PULM-SPECIFICITY-ATYPICAL-L4	atypical organism of low pulmonary specificity, level 4
LOW-PULM-SPECIFICITY-ENV-L4	environmental organism of low pulmonary specificity, level 4
LOW-SPECIFICITY-VIRUS-L4	low-specificity virus, level 4
MNGS-ONLY-NONDOMINANT-NON-TYPICAL-PICKED-OMIT	mNGS-only, non-dominant, non-typical pneumonia organism omitted
MNGS-R0R1-HOST-ONLY-NONPROTECTED-PICKED-OMIT	R0–R1 with host-only support, non-protected: omitted
NON-CORE-RESP-VIRUS-PICKED-OMIT	non-core respiratory virus omitted from picks
NONPULMONARY-CULTURE-CONTEXT-ONLY	culture at a non-pulmonary site is context only
ORAL-ASPIRATION-FLORA-L3	oral/aspiration flora representative retained at level 3
ORAL-ASPIRATION-FLORA-MNGS-ONLY-NONDOMINANT-L4	mNGS-only, non-dominant oral flora, level 4
RARE-YEAST-HOSPITAL-MNGS-L3	rare opportunistic yeast with hospital and mNGS support, level 3
RARE-YEAST-MNGS-ONLY-CONTEXT	rare yeast with mNGS only: context
RELATED-REPRESENTATIVE-HOSPITAL-CONTEXT	related-organism hospital context supports a representative
RESP-COMMENSAL-L3	respiratory commensal retained at level 3
RESP-COMMENSAL-L4	respiratory commensal, level 4
SAME-REPRESENTATIVE-GROUP-PICKED-OMIT	one representative per taxonomic group; others omitted
SKIN-ENV-YEAST-L4	skin/environmental yeast, level 4
SKIN-LOW-SPECIFICITY-BACTERIA-L4	skin flora of low specificity, level 4
TRACE-PROTECTED-WATCHLIST-PREFERRED	trace-read protected pathogen kept on a watch-list, not picked
WATER-ENV-GNB-L4	water-borne environmental Gram-negative, level 4
WATER-ENV-LOW-SPECIFICITY-PICKED-OMIT	water-borne organism of low specificity omitted from picks
WEAK-MOLD-HOST-ONLY-PICKED-OMIT	weak mould signal with host-only support omitted

Table S4: **Evidence-module thresholds used by R5.** Sources: RAG_re/config/default.json, RAG_re/config/formal_merge_full_a.json, RAG_re_casefit/config/default.json, RAG_re_clinical/config/clinical_v1.json, OBER_patient_delivery/generate_r5_final_decisions.py.

Module	Setting
A retrieval	PubMed E-utilities; query "organism"[Title/Abstract] AND (site terms in title/abstract); publication cutoff 2026-06-30; at most 10 articles; 0.34 s between requests; literature also retrieved for upstream picks (<code>literature_on_baseline_picked = true</code>).
A judgement	support requires exact species, human clinical evidence, target-site match and a valid evidence span (≤ 20 contiguous words); positive when ≥ 3 valid supports among ≥ 5 judgeable articles; otherwise abstain.
A1 case-fit	per-article verdict strong / partial / mismatch / insufficient (Methods); aggregation thresholds in config: ≥ 2 strong for high, ≥ 1 strong or ≥ 3 partial for moderate, ≥ 5 judgeable articles. R5 uses strong ≥ 2 or strong+partial ≥ 7 .
B	absolute axis: M1/M2 or R3/R4; relative axis: dominance $\geq D2$, or priority ≤ 1 with percentile ≥ 0.8 ; site alignment required; NTC ratio and RPM gates not required (<code>require_ntc_ratio = false</code> , <code>require_rpm = false</code>).
C	positive modules: culture, FilmArray, FilmArray/galactomannan, targeted molecular, molecular microbiology, PCR, antigen, galactomannan; grades C3 / C2 / C1 / C0 / CNEG as in Methods; respiratory <i>Candida</i> with culture only $\rightarrow$ C1.
D	pass / guarded / block / unknown; reported, not gated.
Temporal	window 3 days before and after the anchor (research arms only).
R5	picked OR C3 OR ((A1_strong ≥ 2 OR A1_match ≥ 7) AND site_aligned AND (B_absolute OR B_relative)); rule id R5_A1_UNION_AND_B_LOOSE.

Table S5: **Language models, prompts and configurations by stage.** SHA-256 prefixes of the files in the code repository at the time of writing (2026-09-20). [\[Confirm the model identifiers and reasoning settings used for the reported runs from the run manifests; confirm the formatter and vision models.\]](#)

Stage	Model / setting	File	SHA-256 (prefix)
Formatter (raw → standardized JSON)	[model]	upstream/core/llm_parser.py	–
Host/laboratory agent	[GPT-5]	agents/prompts/CBC_Lab_underlying_prompt.txt	0929360cfb5d
Culture agent	[GPT-5]	agents/prompts/culture_agent_prompt.txt	e1631735b119
FilmArray and GM agent	[GPT-5]	agents/prompts/filmarray_GMtest_prompt.txt	2a092516ff13
Imaging agent	[GPT-5]	agents/prompts/image_agent_prompt.txt	b0ea51281d8f
Molecular agent	[GPT-5]	agents/prompts/molecular_microbiology_prompt.txt	802ccf52ac71
mNGS-to-specimen	[GPT-5 or deterministic v4]	agents/prompts/mNGS_to_specimen_prompt.txt	643dcf157308
Deterministic summary	rules v4	tools/build_deterministic_summary.py	–
Scorer	rules v19 (reported) / v20 (current)	tools/deterministic_mngs_max_scorer.py	–
Missed-candidate review	[GPT-5]	prompt in tools/review_missed_mngs_candidates.py	–
Article judge (A)	GPT-5.6 Luna, reasoning low, 1,200 tokens	RAG_re/prompts/article_support_v1.txt; config/default.json; formal_merge_full_a.json	af4178902c50; bc123c20c743; b3ca56742dce
Case-fit (A1)	GPT-5.6 Luna, reasoning low, 1,400 tokens	RAG_re_casefit/prompts/article_casefit_v1.txt; evidence_span_repair_v1.txt; config/default.json	e495aef60640; f6f15fec9d9; a9f387850b01
Clinical rules (B, C, D)	clinical_v1	RAG_re_clinical/config/clinical_v1.json	41121351788c
R5	rule R5_A1_UNION_AND_B_LOOSE	OBER_patient_delivery/generate_r5_final_decisions.py	–
Rationale	GPT-5.6 Luna, reasoning low	generate_llm_rationales.py	–
Direct prompting	gpt-5.6-sol/luna/terra, gpt-6-astra; reasoning medium; 4,000 tokens	LLM_test/direct_raw_benchmark/prompt_v1.txt	961087b8acdf

Supplementary Notes

Supplementary Note 1. Missed-candidate safety-review prompt

The review prompt is embedded in `upstream/tools/review_missed_mngs_candidates.py` (`build_prompt`). Its operative content, paraphrased: the model is a safety reviewer for suspected pulmonary infection and may not replace the deterministic picks; it receives the review queue with rank, reads, tiers, dominance, level and guardrail reasons, the deterministic output, the hospital-side summary and the mNGS-to-specimen classes; it returns four categories (`review_high_priority`, `review_context_needed`, `review_low_specificity`, `omitted_with_reason`); it must not upgrade for immunocompromise or protected status alone; respiratory *Candida* and generic yeast evidence is low-specificity unless same-species invasive evidence exists; different *Candida* species are never aliases; CMV, HSV and VZV are flagged as reactivation unless primary pulmonary disease is supported and EBV is never a pulmonary candidate; very high herpesvirus reads with a non-dominant pattern and host-only support remain context-needed; oral and aspiration flora are grouped into strict anaerobes, broad commensals and atypical oral organisms with tier rules for each; water-borne Gram-negatives, GI-only pathogens and coagulase-negative staphylococci or corynebacteria have explicit demotion rules; and the output is a JSON object with Traditional-Chinese rationale fields. [\[Reproduce the prompt verbatim \(with permission\) or provide it as Supplementary Data.\]](#)

Supplementary Note 2. Direct-prompting baseline prompt

File `LLM_test/direct_raw_benchmark/prompt_v1.txt` (SHA-256 prefix 961087b8acdf), original in Traditional Chinese. English rendering:

You are an infectious-diseases clinical reviewer. Using only the single patient's data provided here, decide which microorganisms are most likely the clinical causative pathogens of the current pulmonary or lower-respiratory-tract infection episode.

This is the Direct-Raw baseline: do not use external search, PubMed, RAG, AETIA, any candidate pool or any earlier model result. The raw mNGS list may contain contamination, colonization, background organisms, low-level detections and organisms unrelated to the site of infection; "detected" does not mean "pathogen". Judge from the mNGS reads and RK/NTC codes, the specimen site, cultures, molecular tests, imaging, host status and timing. Respiratory *Candida*, low-specificity environmental organisms, skin flora and latent or reactivated viruses must not be listed as the infection source merely because they were detected. A positive result from a non-pulmonary specimen must not be treated as a pulmonary source without a pulmonary link. An empty array means "no record available", not a negative test. You may select zero, one or several pathogens; do not force an answer. Put only the organisms you finally judge to be the infection source in `predicted_pathogens`; if evidence is insufficient, put them in `excluded_detected_organisms` or `limitations`. Name pathogens at species level; do not merge different *Candida* or other species. Output must follow the given JSON schema; supporting evidence, counter-evidence, reason and limitations should be short and auditable, and hidden reasoning must not be output.

Supplementary Note 3. Case examples

[Two anonymized cases from the January 2026 series (main text, Results): (1) *Klebsiella pneumoniae* selected with 68,569 mNGS reads, bilateral consolidation with pleural effusions, rising CRP and worsening hypoxaemia, with *Candida albicans* excluded as colonization; (2) *Candida tropicalis* selected on multiple positive blood-culture sets and high BALF reads (C3 direct path). Replace with the delivered R5 records: selected organisms, rule paths, module results (A/A1 counts with PMIDs, B axes, C grade, D state), the audited rationale, and the reference labels. Obtain approval for case-level presentation.]

Supplementary Note 4. Version boundaries and replay

The [55] reported R5 decisions were generated from upstream outputs scored with rule version v19 (`mngs_deterministic_level_only_v19_low_specificity_taxonomy`). The v19 scorer was recovered and verified to reproduce all [55] saved scorer outputs from the saved summaries and ranked mNGS. The current scorer v20 (`mngs_deterministic_level_only_v20_central_taxonomy_profile`) moves the taxonomy profiles to a central module and removes a retired language-model scoring branch; replayed on the same inputs it reproduces the picked set of all [55] patients and the candidate names in all [55], changes candidate levels in [15] patients ([15] shared-candidate level changes in total) and the pick order alone in [5], and reproduces the merge-stage picked sets and review-tier membership in all [55]. The clinical-summary replay used the current patient directories rather than a proven frozen agent snapshot; three patients differ in in-hospital evidence between the reconstructed and the saved summaries. None of these differences alters the reported selections. Source: `docs/UPSTREAM_REPLAY_SUMMARY.json`, `docs/UPSTREAM_RECOVERY.md`, `docs/VERSIONS.md`.

Supplementary Note 5. Evaluation outputs to be added

[Per-patient tables for the 55- and 41-patient cohorts (predicted and reference sets per method; TP/FP/FN); per-hospital metrics; macro metrics, any-hit, all-hit and exact-set-match rates; the list of the 9 rescued organisms with rule paths and correctness; error analysis of false negatives (absent from report, below tier, guardrail, review tier not carried) and false positives (upstream pick versus rescue); confidence intervals.]